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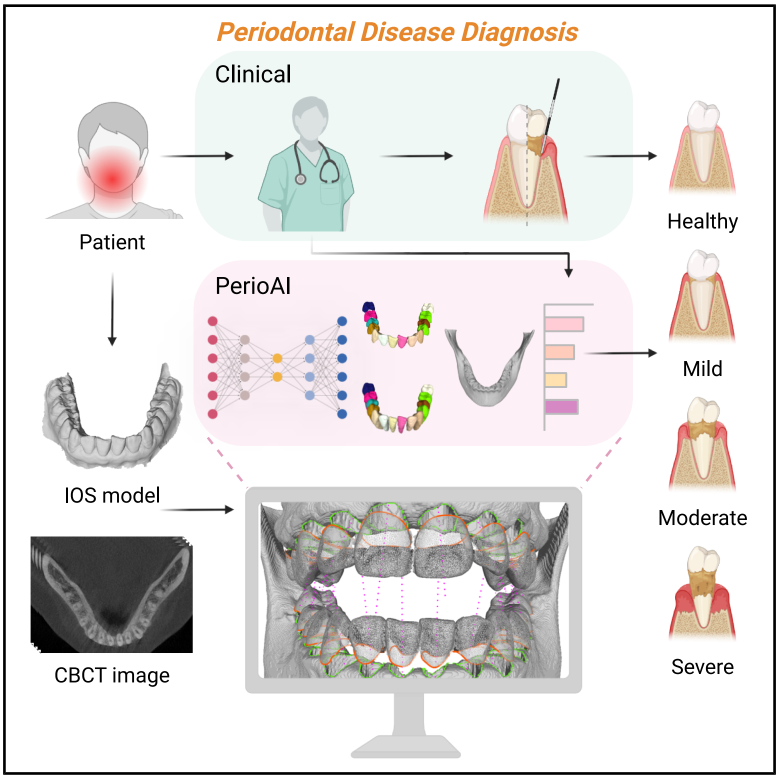
PerioAI: A digital system for periodontal disease diagnosis from an intra-oral scan and cone-beam CT image

Graphical abstract



Highlights

- PerioAI enables automated and non-invasive periodontal diagnosis
- It replaces manual probing by directly measuring GBD using IOS and CBCT data
- A full-stack pipeline integrates segmentation, multimodal fusion, and measurement
- Proven on 2,507 patients, PerioAI enhances clinical workflow and precision

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In brief

Tan et al. present PerioAI, an automated system for precise and non-invasive periodontal diagnosis. By integrating intra-oral scans (IOSs) with CBCT images, PerioAI provides gingiva-bone distance (GBD) and soft and hard tissue information, enhancing diagnostic accuracy and treatment decision-making.

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PerioAI: A digital system for periodontal disease diagnosis from an intra-oral scan and cone-beam CT image

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SUMMARY

Periodontal disease diagnosis and treatment planning are critical for preventing bone and tooth loss. Clinically, dentists manually measure periodontal pocket depth with probes while integrating bone structure from imaging to assess periodontal status, a process that is subjective, invasive, and cognitively burdensome. Here, we propose PerioAI, an accurate, automatic, and non-invasive system that directly measures the gingiva-bone distance (GBD) and provides soft and hard tissue information digitally. PerioAI is a full-stack process comprising four key components: intra-oral scan (IOS) segmentation, cone-beam computed tomography (CBCT) image segmentation, multimodal data fusion, and digital probing measurement. We evaluated PerioAI on multicenter cohorts comprising 2,507 patients. Outstanding IOS and CBCT segmentation performances ensure accuracy throughout the full-stack process. Moreover, digital probing achieves remarkable precision with only 0.040mm error. This approach has the potential to substantially improve clinical workflows in periodontal disease management, offering a more precise, patient-friendly method for diagnosis and treatment decision-making.

INTRODUCTION

Periodontal disease, a pervasive global health issue, results in the progressive and largely irreversible deterioration of the gingiva and alveolar bone, ultimately leading to bone and tooth loss.[1] The Global Burden of Disease Report indicates that 796 million (95% uncertainty interval, 671 to 930 million) suffer from severe periodontitis.[2] Alarmingly, its total burden is increasing annually. A notable challenge in addressing this condition is its asymptomatic presentation, especially in the early stage, which often leads to a delay in seeking therapeutic interventions. Consequently, many patients remain undiagnosed until the disease progresses to a more critical stage.[3,4] This underscores the urgent need for systems that *not only* confirm the presence of periodontal disease *but also* provide concrete guidance for subsequent therapeutic actions.[4]

In clinical settings, periodontal probing remains the primary screening tool for detecting periodontal disease.[5,6] As shown in Figure 1A, the probing technique fundamentally involves inserting a calibrated probe alongside the tooth, beneath the gum line, to measure the depth of the sulcus or pocket. This depth, termed periodontal pocket depth (PPD), indicates

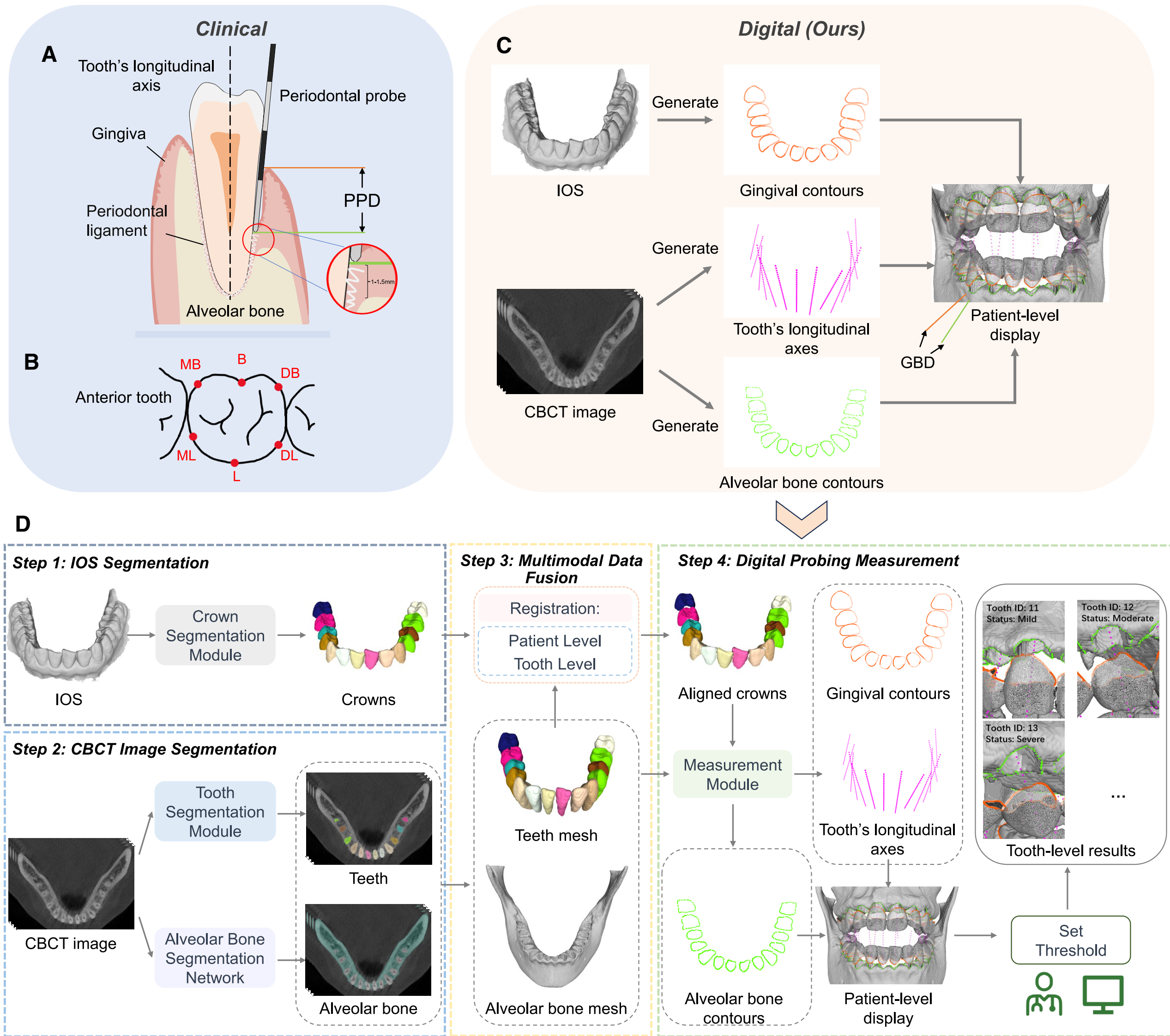

Figure 1. Introduction to clinical probing and PerioAI

(A) Periodontal structure and periodontal pocket depth (PPD) measurement.

(B) Positions of PPD measurement.

(C) Our digital measurement and the gingiva-bone distance (GBD).

(D) Overview of our proposed digital measurement system. Step 1: the crown model is obtained from the IOS by the IOS segmentation module. Step 2: the individual teeth and alveolar bone are segmented from CBCT image by the tooth segmentation module and alveolar bone network, respectively. Step 3: the multimodal data fusion module to align the crown, teeth, and alveolar bone by registration. Step 4: digital probing measurement to measure the GBDs, ensuring the following tooth-level diagnosis.

periodontal health status, where shallower depths generally suggest healthier gums. In contrast, deeper pockets often indicate inflammation and potential periodontitis. Despite its simplicity, this technique has notable limitations. First, the accuracy and reproducibility of clinical measurements can be influenced by manual factors such as dentist experience, the disease status of the individual tooth, probe position, and potential rounding errors.[7–9] Moreover, this technique only records a limited number of points, typically 6 sites,[6,10] around a tooth, making it challenging to fully assess the periodontal condition of the tooth (Figure 1B). Furthermore, the invasive nature of the procedure can cause some degree of patient discomfort during the assessment.

To improve the inadequacy of clinical probing, recent studies have explored digital management methods using imaging images, like panoramic images, intra-oral scans (IOSs), and cone-beam computed tomography (CBCT) images. Many works focus on detecting or calculating alveolar bone loss from

(legend on next page)

panoramic images,[11–21] suitable for initial and periodic screening. However, it only provides two-dimensional projections, making it difficult to accurately determine distorted structures, which may underestimate the depth and structure of intrabony defects.[22] Additionally, these methods cannot indicate another critical diagnostic indicator of periodontal disease progression and treatment needs, i.e., the PPD, as shown in Figure 1A. The second group of approaches[23–25] utilize IOSs or CBCT images, which provide 3D surface or volumetric representations of crowns, teeth, and alveolar bones, exploring visualized 3D models to guide the surgical plan.[26–28] Nevertheless, these methods belong to respective parts of the whole diagnosis process and heavily rely on subsequent hand-made measurements.

The aforementioned technologies can independently capture two critical diagnostic parameters: (1) the status of soft tissue inflammation, as indicated by probing depths, and (2) the quantitative assessment of bone loss through radiographic imaging. However, effective treatment planning requires the integration of these two biological dimensions. Relying on clinicians to mentally synthesize these disparate data streams imposes a significant cognitive burden on them.

In this study, we present an AI-driven system for automated periodontal diagnosis, which integrates multimodal dental data including IOSs and CBCT images (Figure 1C). Our system, named PerioAI, *not only* simulates existing clinical diagnostic workflows to directly calculate gingiva-bone distance (GBD) *but also* provides visual information regarding soft tissue morphology from IOS and the severity of bone loss from CBCT images. This integration enables an accurate diagnosis of periodontal diseases, particularly in the early stages, and facilitates the design of tooth-level treatment, providing guidance for treatment plan. PerioAI was trained and evaluated on a large-scale multicenter dataset comprising 2,507 patients with various combinations of data, including IOSs, CBCT images, and clinical probing records. Our system is full-stack, consisting of four main components (Figure 1D): an IOS segmentation module for crown prediction, a CBCT image segmentation module for teeth and alveolar bone extraction, a multimodal data fusion module for aligning segmented crowns from the IOS with teeth from the CBCT image, and a digital probing measurement module for inferring gingival and alveolar bone contours along the tooth's longitudinal axes. Finally, PerioAI provides tooth-level diagnostic results based on the calculated GBDs and a predetermined classification threshold. In contrast to traditional manual probing methods, our approach offers a digital, reproducible, and less invasive alternative for periodontal disease diagnosis. Users only need to input the IOS and CBCT images, and PerioAI automatically outputs GBD measurements and bone and soft tissue information for tooth-level diagnostic evaluation.

To systematically investigate the effectiveness of PerioAI, we design various tasks for evaluating full-stack digital dental data process and digital probing analysis. Our experiments show outstanding segmentation performance on both internal and external testing datasets, i.e., by achieving an average Dice score (DSC) of 96.8% for crown segmentation from IOSs, 95.9% for tooth segmentation, and 97.2% for alveolar bone segmentation from CBCT images in the internal testing dataset. Additionally, PerioAI can achieve *not only* a remarkable accuracy of only 0.040mm error in digital probing measurement *but also* comparable digital measurements as clinical probing measurements, supporting early diagnosis and enhancing treatment decision-making.

RESULTS

Dataset and evaluation metrics

In this study, we collect a large-scale dataset from the public Teeth3DS dataset[29] and multiple participant centers in China to build a periodontal disease diagnosis system. The participant centers include the Stomatological Hospital of Chongqing Medical University (CQ-hospital), the First People's Hospital of Hangzhou (HZ-hospital), the Ninth People's Hospital of Shanghai Jiao Tong University (SH-hospital), and four dental clinics. All the data obtained from centers involved patients receiving routine clinical care, such as orthodontics, dental implants, restoration, and periodontal disease treatment. The labels (e.g., crowns, teeth, or alveolar bones) of IOSs and CBCT images are carefully annotated and validated by expert raters according to a standardized protocol described in Method Details. Detailed information about the dataset can be found in Figure 2 and Tables S1 and S2. Specifically, we divide all data we collected into internal and external sets. Then, these data are systematically categorized into three categories, aligning with the distinct modules of our digital measurement system, i.e., IOS segmentation, CBCT image segmentation, and digital probing measurement (Figure 2B). In each category, we randomly selected data for training and testing of the corresponding tasks (Figure 2A).

We employ a set of metrics to evaluate the accuracy of the periodontal disease diagnosis system. For the IOS segmentation, we assess accuracy using the DSC (%), mean intersection over union (mIoU, %), and tooth localization accuracy (TLA, %) based on ISO standards. The DSC quantifies the spatial overlap between the segmentation result (R) and the ground-truth (G), defined as $DSC = \frac{2|R \cap G|}{|R|+|G|}$. The mIoU measures segmentation performance by calculating the ratio of the intersection to the union of the segmented and ground-truth labels. TLA is the normalized

Figure 2. Detail of the dataset and results of the full-stack digital dental data process

(A) Data partition in experiments.

(B) We divide data into different parts based on the data types, modules, and tasks.

(C) Our IOS segmentation method outperforms the other seven methods in terms of Dice score on both internal and external datasets. Error bars indicate the 95% confidence intervals (95% CIs).

(D) Tooth and bone segmentation based on CBCT images outperforms HMG-Net in terms of Dice score for both internal and external datasets. Error bars represent the 95% CIs.

(E) Representative caries case demonstrates the high accuracy of the PerioAI system's segmentation module, the registration module's ability to minimize cumulative errors, and the overall robustness of the system's measurements.

Euclidean distance between the ground-truth teeth centroid and the closest localized teeth centroid. For CBCT image segmentation, our evaluation incorporates the DSC and average surface distance (ASD, mm). The ASD metric, meanwhile, gauges the difference between the segmentation result (R) and the ground truth (G). For data registration, we employ both fitness and inlier Root Mean Square Error (inlier RMSE, mm) to assess the quality of alignment. Here, fitness evaluates the proportion of successfully matched point pairs between dental crown and tooth, while inlier RMSE measures the accuracy of these matched point pairs by calculating the root-mean-square error of the distances between matched points. A high fitness score coupled with a lower inlier RMSE value indicates more precise alignment. Additionally, for digital probing measurement, we use the Euclidean distance to measure the distance between the gingival contour and the alveolar bone contour.

Full-stack digital dental data process: IOS segmentation

As a key component of our full-stack digital dental data processing pipeline, the IOS segmentation module is evaluated here. In particular, we compare it with several established approaches, such as PointNet,[30] PointNet++,[31] Point Cloud Transformer (PCT),[32] TSeg-Net,[33] SGTNet,[34] DBGANet,[35] and WS-TIS.[36] To ensure a fair and reliable comparison in our experiments, we standardize the output across all methods to 33 classes, encompassing 32 tooth categories and 1 gum background, to maintain consistency. Figure 2C presents a comparison of DSC values. A more comprehensive analysis, which includes the comparison results of DSC, mIoU, and TLA, is provided in Table S3. Our proposed method performs better than other methods, with DSC values of 96.8% and 96.5% on the internal and external testing sets, respectively. These outcomes underscore the effectiveness of our unique strategies, which include the model alignment module, a semantic segmentation module, an instance segmentation module, and a label optimization module. The competing methods impose stringent requirements on the direction of the input grid, necessitating a unified input direction during testing, which presents challenges for practical applications. Unlike competing methods that either input the IOS directly for multiclass semantic segmentation or rely solely on centroid information for crown localization, our method effectively combines the results of semantic and instance segmentation. This integrated approach facilitates a more accurate characterization of each crown, resulting in robust individual crown segmentation. Furthermore, the label optimization module markedly enhances tooth labeling accuracy, underscoring the advanced capabilities of our method.

Full-stack digital dental data process: CBCT image segmentation

As another important component of our full-stack digital dental data process, we evaluate the CBCT image segmentation module here, which targets segmentations of both teeth and alveolar bone. We compare our method with two baselines: HMG-Net[37] (for both teeth and alveolar bone segmentations) and MWT-Net[38] (for tooth segmentation alone). Figures 2D and S1 and Table S4 show quantitative and qualitative results for tooth and alveolar bone.

Specifically, in terms of tooth segmentation, our method exhibits superior performance, achieving the average DSC score of 95.9% and 92.2% on the internal and external testing sets, respectively (Figure 2D; Table S4). We also provide a visual comparison of missing segmentation, apical segmentation, and tooth segmentation with metal artifacts, as shown in Figures S1A–S1D. In Figure S1A, the HMG-Net completely ignores the tooth with caries as it adopts a strategy of first detecting and then segmenting. Figure S1B shows a normal tooth, where it can be observed that the HMG-Net exhibits inadequate segmentation in the root region. Conversely, our method excels in accurately segmenting the complete morphology of these teeth, including the tooth root apex. Additionally, the metal artifacts introduced by dental fillings (Figure S1C) or metal crowns (Figure S1D) significantly affect the quality of segmentation. HMG-Net performs poorly in segmenting these types of teeth, while our method can segment individual teeth robustly and accurately. All these results highlight the effectiveness of our proposed bottom-up (i.e., segment-then-detect) approach, which integrates a semantic segmentation network with a marker-based watershed algorithm.

In terms of alveolar bone segmentation, our method also achieves significant improvement over HMG-Net, as evidenced by Table S4, by achieving DSC scores of 97.2% and 95.9% on the internal and external testing sets, respectively. In addition, Figure S1 shows an example of bone loss. Our method still successfully delineates the boundaries of the alveolar bone, which is of significant importance for subsequent measurements.

Full-stack digital dental data process: Error analysis in segmentation, fusion, and measurement

Accurate segmentation is crucial as it directly impacts the precision of subsequent measurements, and error propagation in each segmentation step could potentially have a cumulative effect on the final diagnosis. To analyze these issues, we designed a comparative experiment in which the IOS segmentation and CBCT segmentation modules in the PerioAI system were replaced with the best-performing approach (excluding our method) identified in the module comparison experiments (see Tables S3 and S4). This experiment enables us to assess the registration results and the final GBD measurement outcomes, thereby verifying the accuracy of our system. An external validation set consisting of data from 56 patients at clinic 4 was used, and the specific results are presented in Table 1.

Two key findings can be drawn from the table. First, the segmentation results of the two modalities significantly influence the effectiveness of registration. A comparison of the four methods presented shows that the optimal registration outcomes can only be achieved when segmentations of both modalities are performed at their highest quality, given the same registration method. For instance, the PerioAI system attains the highest registered fitness score of 0.971 and a relatively low inline RMSE of 0.211mm. In contrast, the combination of PCT_HMG leads to the poorest registration outcomes. This suboptimal performance can be attributed to the necessity of tooth-level registration; inaccuracies in tooth morphological segmentation or misclassification can introduce deviations in the results. Second, our global-to-local registration strategy, when combined with the

Table 1. Comparative analysis of different segmentations for data fusion and GBD measurement

	IOS segmentation		CBCT image segmentation		Registration		Digital probing measurement	
Methods	PCT	Ours	HMG-Net	Ours	Fitness	Inlier RMSE	Tooth average distance error (mm)	Std. (mm)
PCT_Ours	✓			✓	0.873	0.198	0.457	1.136
PCT_HMG	✓		✓		0.843	0.199	0.881	1.432
Ours_HMG		✓	✓		0.918	0.207	0.419	1.201
PerioAI		✓		✓	0.971	0.211	0.037	0.366

best segmentation module, effectively minimizes the risk of cumulative errors, resulting in superior measurement outcomes. The PerioAI system demonstrates a minimum average distance of 0.037mm and a minimum standard deviation of 0.366mm, which are significantly lower than those observed in other comparative experiments. In summary, in conjunction with previous experimental findings, these results indicate that the integration of the optimized segmentation module facilitates excellent registration, and the error reduction achieved during the registration process contributes to superior measurement results.

To further illustrate the robustness of our system, we selected a representative case of caries, as depicted in Figure 2E. In this patient, tooth 16 exhibited significant caries, which posed challenges for both IOS segmentation and CBCT image segmentation. However, our approach demonstrates superior performance in accurately delineating the extent of caries compared to other methods. First, through patient-tooth level registration, our PerioAI system achieves optimal matching results, as evidenced by the lowest average Harsdorff distance. Additionally, our method yields better segmentation results, enhancing the accuracy of the measurement outcomes. In contrast, the PCT method's incorrect crown segmentation can adversely affect the gingival edge, resulting in inaccuracies in the delineation of the gingival contour, as illustrated in the first two lines of the figure. Specifically, misalignment between teeth 16 and 17 can lead to blurred gingival contour for tooth 16, resulting in partial overlap with tooth 17. Furthermore, inadequate tooth segmentation by the HMG-Net method can result in inaccuracy in the longitudinal axis and registration position of the teeth, ultimately leading to discrepancies in measurement direction and contour point positioning.

Digital probing analysis: GBD measurement in digital space

To verify the clinical applicability of our AI system for accurate diagnosis of periodontal disease, we conduct a two-part digital probing analysis: GBD measurement comparison in digital space and comparison with the clinical probing method. First, we validate the accuracy of GBD measurement in digital space by creating two scenarios using an external dataset consisting of 147 patients (3,983 teeth): (1) the shortest distance (Short.) and (2) the distance along the tooth's longitudinal axes (Ours).

Figure 3 visually illustrates different contours in these digital measurement methods, Figure 3A, 3B, and 3D-3G show predicted tooth crowns, predicted alveolar bone, and gingival contours ($Pred_g$) from predicted crowns. In Figure 3A, the brown contours are the predicted bone contours ($Pred_{bs}$) derived using the shortest distance between the $Pred_g$ and predicted bone. In Figure 3B, the green predicted bone contours ($Pred_b$) are ascertained based on the tooth's longitudinal axis direction. Figure 3C shows the ground-truth annotation for the crown, alveolar bone, gingival contours (GT_g), and alveolar bone contours (GT_b), which are annotated by dental professionals. It can be observed that, in bone contour generation, the $Pred_b$ aligns more closely with GT_b. At the same time, the shortest distance rule, a method inconsistent with clinical practices, leads to multiple gingival sites sharing a single bone position point. Figures 3D–3G provide four detailed examples of digital probing measurement for these two methods. In these typical cases, the number of corresponding points in the $Pred_b$ is much more than that in $Pred_{bs}$, leading to the distance *M0* (our method) being more reasonable than *M1* (Short. approach). Additionally, Table 2 summarizes all quantitative results, including errors between the two methods and the ground-truth. From the table, we can obtain two observations. First, across all tooth types, our measurements are more accurate than those using the shortest distance approach. Second, our method, which involves the tooth's longitudinal axis, aligns with the established clinical probing guidelines, with an average distance error of 0.040 mm on the external testing set.

Digital probing analysis: GBD measurement in clinical space

The traditional method measures PPD with a calibrated probe, which has limitations in accurately reflecting periodontal disease. As depicted in Figures 1A and 1B, the periodontal connective tissues, such as the periodontal ligament, limit the probe tip to about 1–1.5 mm above the alveolar bone, providing only an indirect indication of bone loss. Additionally, clinical measurements only record six specific sites (i.e., Mesial Buccal (MB), Buccal (B), Distal Buccal (DB), Mesial Lingual (ML), Lingual (L), and Distal Lingual (DL)) around each tooth, which may not fully represent dental health. In this section, we simulate the six-site measurement process to demonstrate our system's superiority over conventional clinical probing and automatically identify equivalent sites in the digital space.

Specifically, our study included data from 37 patients with 967 teeth, each with IOS, CBCT image, and clinical probing values. Table S2 presents the demographic characteristics, disease staging, and other relevant variables of the 37 patients included in this study. Notably, the periodontal disease staging in this table is determined by the overall health status of each patient, incorporating comprehensive PPD along with other clinical indicators such as clinical attachment level (CAL) and gingival bleeding. However, the primary focus of this study is to evaluate

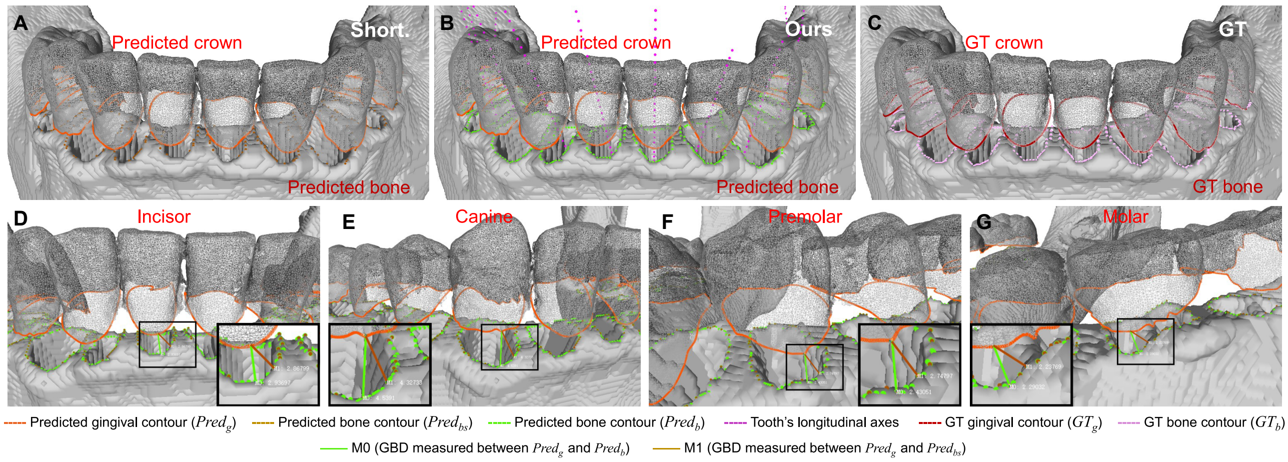

Figure 3. Visualization of different contours in digital measurement and four typical examples of GBD using shortest distance (Short.) and using the tooth's longitudinal axis method (Ours)

(A) Visualization of different contours in the shortest distance (Short.) approach.

(B) Visualization of different contours measured along the tooth's longitudinal axes (Ours).

(C) Detailed view of the GT crown and predicted bone contours.

(D-G) Typical examples of GBD using the shortest distance method (Short.) and the tooth's longitudinal axis method (Ours) for incisor, canine, premolar, and molar.

periodontal disease at the individual tooth level using digital GBD measurements. To achieve this, we adopted a clinician-recommended tooth-level disease classification approach. Specifically, the gingival contour along the occlusal plane was divided into six regions, and the maximum probing depth from each region was extracted as the measurement value for that site, resulting in a total of 5,804 measurement sites. Based on clinical thresholds, we categorized the measurements into three groups: $\leq$3 mm, 4$-$5 mm, and >5 mm, corresponding to mild, moderate, and severe periodontal disease, respectively.

Figures 4A and 4B illustrate the distribution of patient-level disease diagnoses to site-specific probing depth measurements. The results indicate that the majority of probing values in patients with stage 1 periodontal disease are $\leq$3 mm, while the proportion of values measuring 4$-$5 mm and >5 mm increases significantly with the severity of the disease. Figure 4C further explores the relationship between patient-level diagnoses and digital measurement errors. Given that digital measurements tend to be approximately 1.5 mm higher than clinical values—primarily due to soft tissue thickness—an error value close to 1.5 mm indicates better measurement consistency. As depicted in the figure, the median measurement error across different patient-level diagnoses remains close to 1.5 mm, although some outliers are observed in certain patients.

Building on these findings, we conducted a further analysis of the correlations among disease severity, tooth type, and site location. The line chart in Figure 4D displays the distribution of errors between digital and clinical measurements, with peak errors concentrated around 1.5 mm, indicating a strong level of agreement between the two methods across all sites. Figure 4E illustrates the distribution of measurement sites within the three disease categories across different tooth types. Overall, the dataset encompasses a substantial number of sites in the $\leq$3 mm category, while moderate to severe periodontal disease (>3 mm) is predominantly observed in the molar.

Figure 4F further examines the distribution of errors across the six lingual and buccal measurement sites based on tooth-level disease classification and tooth type, revealing four key findings. First, in the mild periodontal disease category ($\leq$3 mm), the median error across all sites is approximately 1.5 mm, with a relatively uniform distribution, indicating stable system performance in this category. This suggests that the system holds promise for the early detection of periodontal disease, facilitating timely diagnosis. Second, in the moderate (4$-$5 mm) and severe (>5 mm) categories, the error distribution exhibits site dependency. Specifically, errors at lingual sites (L, ML, and DL) are closer to 1.5 mm, whereas errors at buccal sites (B, MB, and DB) are slightly higher, suggesting that the system achieves greater measurement accuracy at lingual sites. This may be attributed to anatomical differences in periodontal tissue structure and patterns of bone resorption. Third, in patients with severe periodontal disease, outliers in measurement errors are more frequent, indicating increased discrepancies between digital measurements and clinical probing in advanced disease stages. This phenomenon could be influenced by factors such as probe placement, angulation, tooth hypermobility and operator-dependent subjective judgment during clinical probing. Figure S2 further illustrates potential sources of these errors. Lastly, across all disease categories, molars exhibit measurement errors that are generally closer to clinical values, which may be partially attributable to the distribution of data volume.

DISCUSSION

The persistently high prevalence of periodontitis can be largely attributed to the lack of timely diagnosis and treatment. A key

Table 2. Comparison of error between ground-truth and prediction under different methods, i.e., using the shortest distance (Short.) between gingival contour and bone contour, and the distance along the tooth's longitudinal axis (Ours) between gingival contour and bone contour

	Short.		Ours	
Tooth class	Tooth average distance error (mm)	Std. (mm)	Tooth average distance error (mm)	Std. (mm)
Incisor	0.485	0.176	0.060	0.128
Canine	0.376	0.098	0.010	0.134
Premolar	0.233	0.102	0.010	0.091
Molar	0.295	0.137	0.063	0.062
Average	0.343	0.142	0.040	0.100

challenge is the reliance on manual probing, a skill-intensive technique that necessitates highly trained clinicians, who are scarce and typically available only in a few top hospitals across China. Although expanding the workforce of skilled practitioners could partially alleviate this issue, manual probing remains a labor-intensive and subjective process.[4] While radiographic imaging techniques (e.g., panoramic images) facilitate computer-assisted bone loss assessment, they do not eliminate the necessity for clinicians to mentally integrate probing depth measurements with structural bone information when formulating treatment plans. This cognitive burden further complicates accurate diagnosis and decision-making. Fortunately, the increasing availability of IOS and CBCT,[39–47] which provide detailed three-dimensional visual models, presents a promising opportunity to address these challenges.

In this work, we present PerioAI, a fully automated system designed for diagnostics and support in treatment decision-making. Unlike traditional methods, PerioAI directly measures the GBD in a digital space, integrating soft tissue morphology data obtained from IOSs with the severity of bone loss depicted in CBCT images. This integration facilitates a multidimensional evaluation of periodontal health, effectively bridging the gap between diagnostic measurements and clinical decision-making.

Our system's primary advantage lies in its adherence to clinical diagnostic measurement guidance, providing a comprehensive, fully automated, and non-invasive diagnostic approach. While widely used, the conventional probing approach for assessing periodontal disease has a few limitations. First, the use of a calibrated probe offers only an approximation of periodontal health, constrained by physical characteristics of periodontal tissues. The periodontal ligament, for instance, restricts the probe to reach a point just above the alveolar bone, resulting in an indirect measure of bone loss. Second, by relying on discrete measurements from six sites around each tooth, the method may overlook areas of concern, potentially underrepresenting the overall periodontal condition. These constraints highlight the need for more comprehensive diagnostic tools that are capable of providing a more accurate and holistic evaluation of periodontal health.

PerioAI directly measures GBD in the digital space, providing a comprehensive view of gingival recession and alveolar bone loss, which is essential for effective treatment planning. It operates without manual intervention and requires only IOS and CBCT image inputs, offering patients more comfortable and accurate diagnostic experience. This approach likely improves acceptability and adoption while eliminating the variability introduced by manual examination and different expertise. Another significant contribution of this study is a series of experiments conducted on extensive datasets with various stages of periodontal diseases, showcasing great precision of our system in segmentation and measurement. Our system demonstrates stable crown and tooth segmentation performance on the external testing dataset (e.g., with DSC of 96.5% in crown segmentation and DSC of 92.2% in tooth segmentation). Moreover, we achieve better results on the internal testing set compared to reported performances by competitive methods. Alveolar bone segmentation, leveraging a large dataset of periodontal inflammation cases, shows increased precision with DSCs of 97.2% and 95.9% on the internal and external testing sets, respectively. These outcomes enable the accurate measurement of distances between gingival and alveolar bone contours along the tooth's long axes and provide valuable guidance for periodontal disease diagnosis.

Additionally, our small-scale experiments with 56 data points demonstrate that the global-to-local registration strategy is crucial for mitigating the cumulative effect of segmentation errors. Our results indicate that optimal registration performance is achieved only when both imaging modalities employ the best segmentation methods. For instance, the PerioAI system attained a registration fitness score of 0.971 and an inline RMSE of 0.211mm, evidencing that segmentation errors were effectively controlled during the patient-tooth-level registration process. Notably, this advantage becomes even more pronounced in complex pathological cases, such as severe caries on tooth 16, where precise alignment of the tooth and surrounding structures is critical. Overall, these findings provide compelling evidence that integrating optimized segmentation modules with our advanced registration strategy not only minimizes cumulative error but also substantially enhances the accuracy of digital measurements.

Moreover, the results measured by PerioAI in digital space, comprising 360 points around each tooth, show minimal error of just 0.040 mm compared to the ground truth on the external set. This negligible error, particularly when compared with manual measurement, underscores the exceptional accuracy of our system. Furthermore, the validation of the tooth's longitudinal axis direction is supported by both qualitative and quantitative analyses, indicating that our method yields results *not only* more consistent with clinical knowledge *but also* exhibiting greater accuracy than conventional methods. A typical case is presented in Figure 5A-C, which illustrates the difference between clinical probing results and digital measurements in clinical validation. Unlike clinical measurements, our method directly measures the distance from the gingiva to the bone. By incorporating multiple measurement sites, our approach offers a more comprehensive representation of the overall condition of each tooth, compared to the six sites measured in clinical probing.

In summary, PerioAI has introduced a groundbreaking system that establishes an informational integration entity for 3D

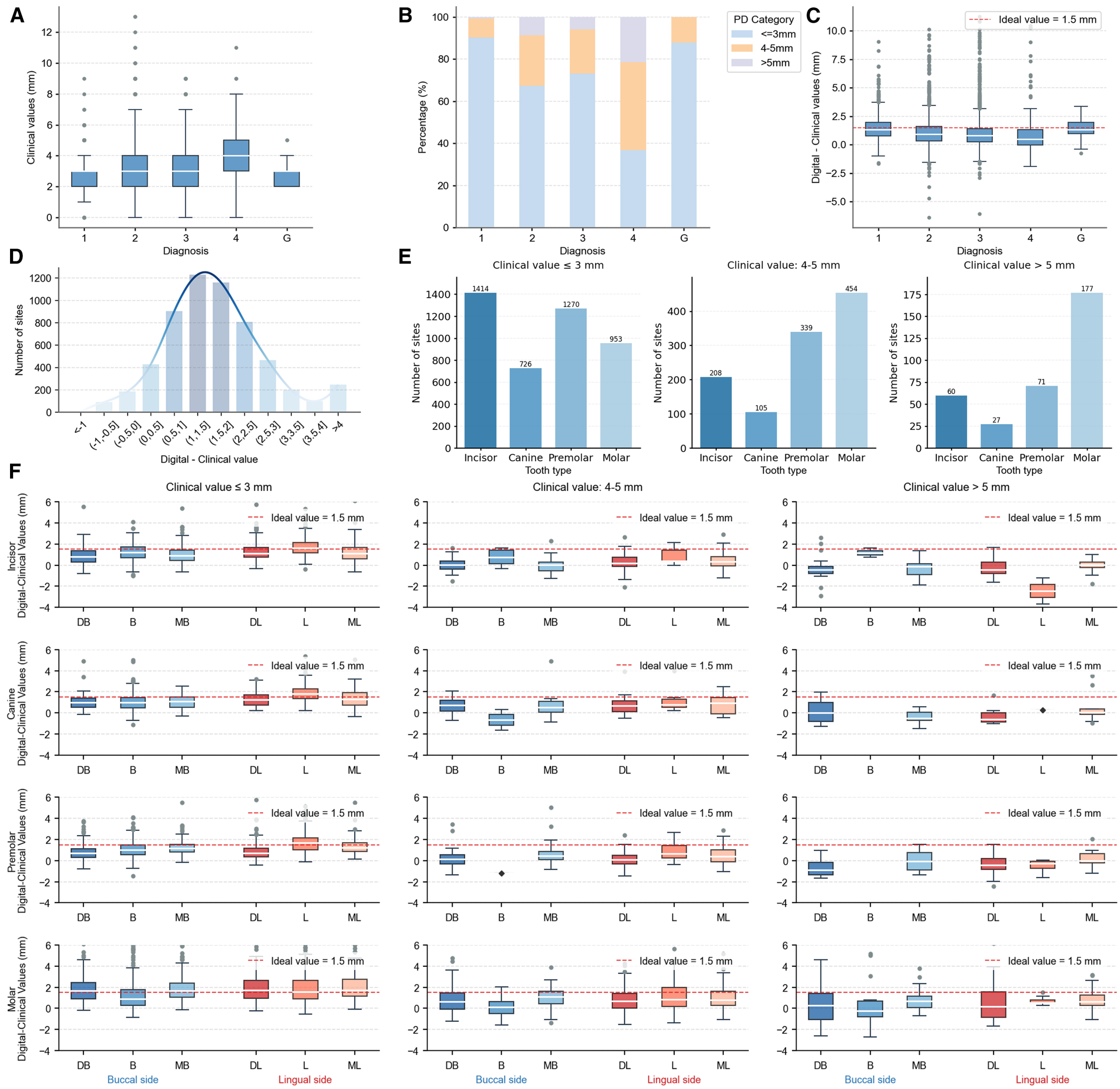

Figure 4. Comparison of digital-clinical values across different clinical ranges, tooth types, and measurement sites

(A) The relationship between patient-level diagnosis and clinical probing values of dental sites. Boxplots show the interquartile range (IQR, 25th–75th percentile), with the median indicated by a line inside the box. Whiskers extend to the most extreme data points within 1.5× IQR from Q1 or Q3. Data points beyond this range are plotted as individual outliers. Diagnosis 1 to 4 are patient level stage I to IV periodontitis while G is gingivitis.

(B) The percentage of three classifications based on clinical probing values of tooth sites in patient-level diagnostic categories.

(C) According to the classification of clinical patient-level diagnosis, comparing the errors of digital measurements and clinical values, the results show that the digital results are approximately consistent with the ideal values. Definitions of boxplot elements are as in (A). The red dashed line indicates the ideal, anatomically determined, measurement agreement between digital and clinical values.

(D) Distribution of differences between digital measurements and clinical values for six sites of all teeth.

(E) The distribution of the number of loci contained in different types of teeth based on the three categories of clinical test values.

(F) The distribution of errors between digital and clinical values for different tooth types (incisor, canine, premolar, and molar) and different sites (buccal side [DB, B, and MB] and lingual sides [DL, L, and ML]) across three clinical ranges (≤3 mm, 4−5 mm, >5 mm). The red dashed line indicates the deviation between the best numerical measurement and the clinical value, the gray circles represent outliers, and the dark gray diamonds denote the measurement error of the unique data points in each category. Definitions of boxplot elements are as in (A).

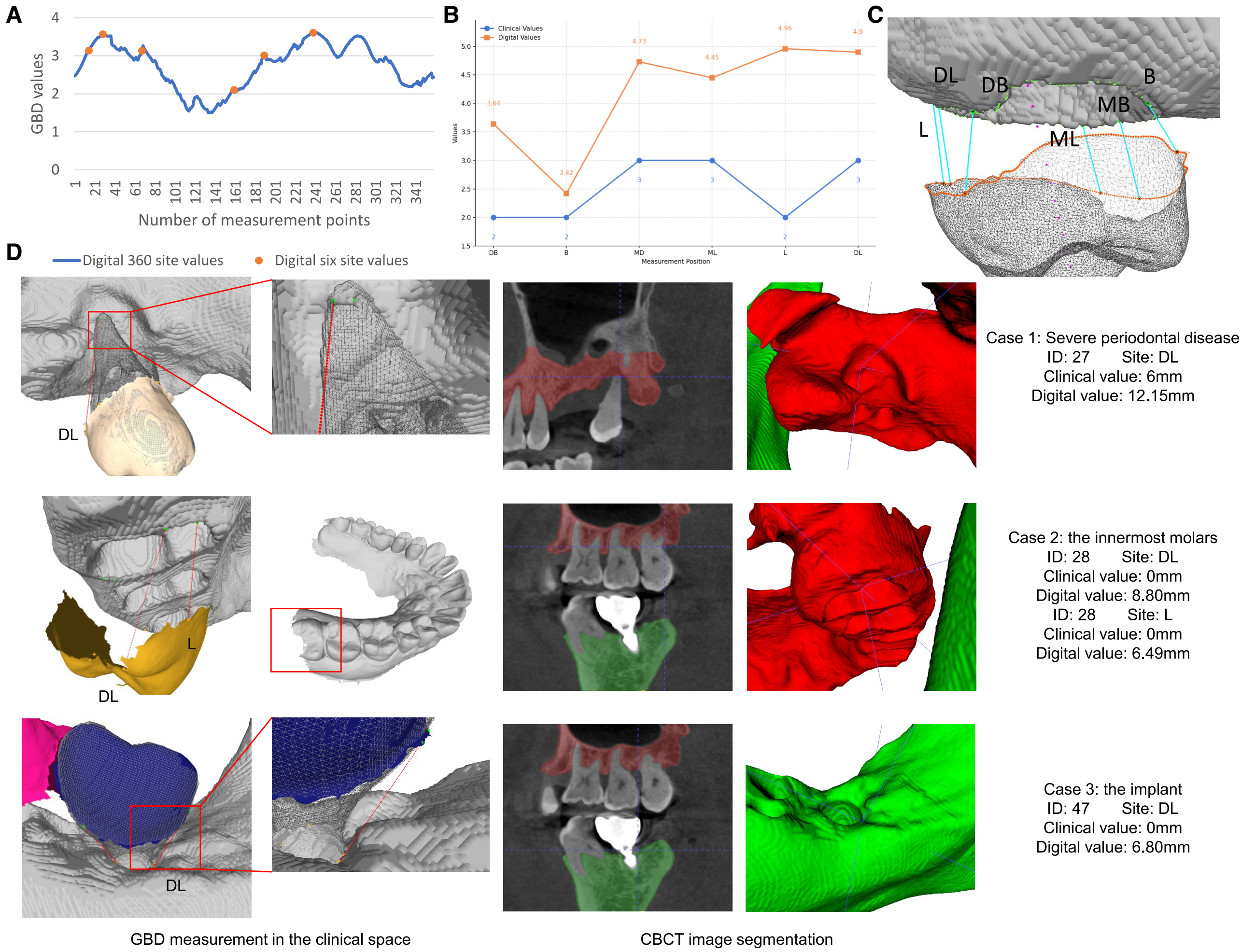

Figure 5. Typical examples of 360-point and 6-point measurements in PerioAI systems and three shortcomings that need improvement
(A) Our digital measurement results include 360 sites for comprehensive diagnosis and 6 sites for comparing with clinical values.
(B) Numerical comparison of clinical and digital measurement results at the six sites.
(C) Visualization of digital measurement.
(D) Insufficient measurement of severe periodontal disease, innermost molars (wisdom teeth), and dental implants.

periodontal diagnostics, encompassing GBD values, soft tissue, and alveolar bone information. We believe that this system will propel further research and serve as a foundational element in AI-assisted periodontal diagnosis and treatment, potentially transforming the landscape of periodontal care.

Limitations of the study

The study implies that the integration of AI with dental medicine holds promise for advancing digital dentistry. Our PerioAI system is capable of learning from the expertise of clinical practitioners, thereby generating consistent results. Moreover, the long-term cost-effectiveness of our proposed system, compared to traditional manual diagnostic methods (e.g., probing), positions it as a viable option for resource-limited areas. Although our work has shown promising results in segmentation, measurement, and clinical potential, it also presents some limitations that necessitate further investigation.

Firstly, our digital measurement approach involves a complex workflow—including IOS segmentation, CBCT image segmentation, multimodal data fusion, and digital probing measurement. Each step introduces the potential for error, and these errors may accumulate throughout the process. Although we designed comparative experiments to analyze error propagation, there remains substantial opportunity for enhancing the overall performance of the system. To address this, we plan to integrate multimodal dental data to jointly segment teeth and crowns, which should further improve accuracy.

Secondly, our system exhibits certain shortcomings when measuring extreme cases. In our analysis of outliers (as shown in Figure 4F and illustrated with three typical cases in Figure 5D), we identified challenges in three specific scenarios: severe periodontitis, the most posterior molar, and implant cases. For teeth affected by severe periodontitis—with marked bone resorption and near-separation of the roots from the alveolar bone—the CBCT-based bone segmentation proved challenging, and the resulting measurement path occasionally intersected the tooth contours. In future work, in addition to improving segmentation performance, we plan to incorporate tooth morphology as a constraint to enhance measurement robustness. For the most posterior molar, discrepancies between clinical and digital measurements can arise from two main factors. First, clinicians frequently overlook probing this tooth. Second, the limited field of view during intraoral scanning may result in an incomplete IOS model, potentially leading to misestimation of GBD. To address these limitations, we intend to integrate tooth morphology information from CBCT into the IOS model, thereby reconstructing the missing anatomical structures and enabling more accurate and comprehensive GBD assessment across the full dentition. For implant cases, although our digital measurements yield results, the lack of an established gold standard remains a concern. Future studies will involve close collaboration with clinicians to better evaluate and refine measurements for these challenging cases.

Third, our work aims to propose a practical method for diagnosing periodontal disease and assisting in treatment planning, which has not yet been widely tested to demonstrate its broad applicability in clinical environments that support periodontal diagnosis and treatment planning. This task requires significant time and effort, and we plan to conduct further clinical validation in the future. Additionally, our current method relies on medical institutions equipped with CBCT and IOS technology. For scenarios that do not meet these device requirements, our goal is to explore more accessible imaging options, such as photos taken with standard cameras, to directly assess periodontal health and generate 3D models of oral tissue for treatment guidance. This will also be the focus of our future work.

RESOURCE AVAILABILITY

Lead contact

Requests for further information, resources, and reagents should be directed to and will be fulfilled by the lead contact, Dinggang Shen (Dinggang.Shen@gmail.com).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- A subset of 600 cases from the publicly available training set of the Teeth3DS dataset was used in this study for IOS segmentation. This dataset comprises 3D intra-oral surface scans and corresponding annotations and was developed for the 3DTeethSeg Challenge at the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) in 2022. Accession information is provided in the key resources table.
- Other datasets are protected because of privacy issues and hospital regulation policies, which take time to resolve and process. Partial internal raw data will be shared by the lead contact upon request.
- The inference code version of this system has been deposited at Zenodo and is publicly available as of the date of publication. DOIs are listed in the key resources table. It should be used for academic research only.
- Any additional information required to reanalyze the data reported in this work paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, M.S.T., D.S., M.T., and Z.C.; methodology, M.S.T., D.S., M.T., Z.C., Y.L., and Y.F.; resources, M.S.T., Y.L., Y.Z., X.W., and H.L.; writing – original draft, M.T., Z.C., and Y.L.; funding acquisition, M.S.T., D.S., Z.C., Y.L., and Y.Z.; supervision, M.S.T. and D.S.

DECLARATION OF INTERESTS

D.S. is an employee of Shanghai United Imaging Intelligence Co., Ltd.; M.S.T., Y.L., X.W., and H.L. are Shanghai Ninth People's Hospital employees; M.S.T., D.S., M.T., Z.C., and Y.L. have a patent application related to this work (202411291185.6).

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Teeth3DS	3DTeethSeg Challenge at MICCAI 2022	OSF: https://osf.io/xctdy/
Software and algorithms		
ITK-SNAP 4.0.2	ITK-SNAP Software	http://www.itksnap.org/pmwiki/pmwiki.php
MeshLab 2023.12	MeshLab Software	https://www.meshlab.net/
HMG-Net	A fully automatic AI system for tooth and alveolar bone segmentation from cone-beam CT images	https://github.com/ErdanC/Tooth-and-alveolar-bone-segmentation-from-CBCT
SGTNet	3D Dental Mesh Segmentation Using Semantics-Based Feature Learning with Graph-Transformer	https://github.com/df-boy/SGTNet
DBGANet	DBGANet: dual branch geometric attention network for accurate 3D tooth segmentation	https://github.com/zhijieL513/DBGANet
Original code about digital periodontal diagnosis system	This paper	https://zenodo.org/records/15297291
Other		
NVIDIA GeForce RTX 4090 GPU	NVIDIA GeForce RTX 4090 GPU	https://www.nvidia.com/en-in/geforce/graphics-cards/40-series/rtx-4090/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Patient

This study included a total of 2507 patients from a public Teeth3DS dataset and multiple participant centers in China. The data partition is shown in Figures 2A and 2B. Specifically, the data analysis for the "digital probing analysis: GBD measurement in clinical space" section was conducted using data from 37 periodontal patients who sought dental care at Shanghai Ninth People's Hospital between December 2021 and May 2022. All participants were adults aged $\geq$18 years, with a balanced distribution of 19 females and 18 males. The demographic characteristics of the participants are summarized in Table S2, which provides detailed information on age, gender, smoking status, and clinical parameters such as the number of teeth, full-mouth bleeding score (FMBS), PPD, and CAL across different stages of periodontitis. All samples used in this study were deidentified and obtained with the properly informed consent of patients and/or their legal representatives, who did not receive compensation. The study was approved by the Institutional Review Board of the Shanghai Ninth People's Hospital (SH9H-2021-T408-3).

METHOD DETAILS

Data annotation protocol

In our study, we have developed the PerioAI system to accurately measure periodontal probing distances in three-dimensional space, thereby assisting clinicians in assessing the severity of periodontitis and making treatment plans. The system is based on extensive CBCT and IOS model data. To ensure high-quality annotations, six experienced clinicians participated in the process. Five junior doctors annotated the data, with one senior expert reviewing all annotations. We also established a detailed standardized protocol to maintain consistency across different annotators.

The details of the annotation process are as follows. For IOS data, clinicians marked the crown edges using the color and texture of tooth and gum as references. For CBCT data, clinicians employed threshold segmentation plugins in 3D Slicer and ITK-SNAP software, manually correcting the results to annotate teeth and alveolar bone. The annotated voxel maps were subsequently converted into mesh representations. Following this, CBCT and IOS data were manually registered using MeshLab software, from which intersection points at the junction of tooth and bone meshes were extracted to delineate the bone contour. After obtaining the crowns, we extracted boundary points and applied spline interpolation to complete the gingival contours, utilizing a specified number of points (i.e., 360 points in this study). Finally, clinicians paired each gingival point with its corresponding bone point, following the direction of clinical probing. These paired points were saved in CSV files for further analysis.

IOS segmentation

We employ our previous crown segmentation network[33,48] to segment and label crowns accurately, including a rotation module, a tooth instance segmentation module, a tooth semantic segmentation network, and a label optimization module. The details are shown in Figure S3A. Considering that the IOSs captured by different devices have different coordinate standards (i.e., different orientations), we first use a rotation network to align them to the same coordinate system with the same orientation. Then, we use the tooth instance segmentation module to obtain accurate individual teeth. Specifically, we predict the tooth centroids and crop the aligned model into a set of patches based on these centroids. A distance weight (w) is calculated for each point to mark the tooth that needs to be segmented. Additionally, we calculate the curvature features (f) from the results of a binary gingival separation network to delineate more accurate crown boundaries. Each point in a patch is represented by [x, y, z, n_x, n_y, n_z, f_i, w_i] and then sent to the instance segmentation network to divide the entire IOS into binary tooth labels. Following this, we utilize a tooth semantic segmentation network to find the optimal label for each tooth instance. We combine the instance segmentation (object-level) with semantic segmentation (point-wise) together for more precise segmentation and rough labeling results. Finally, we employ a label correction module, which uses a cost ranking algorithm and combines it with arch curves to minimize the problem of label inversion and identical labels.

CBCT image segmentation

Recognizing that the essential components for GBD measurement, i.e., teeth and alveolar bone, are extracted from CBCT images, we develop a CBCT image segmentation methodology consisting of a tooth segmentation module and an alveolar bone segmentation module.

Existing tooth segmentation methods,[37,38,49,50] which often employ a "detect-then-segment" strategy, can struggle with segmenting tiny teeth or accurately distinguishing closely neighboring teeth, particularly in cases involving teeth with metal artifacts. Precise segmentation in these challenging scenarios is crucial for periodontal disease diagnosis. Therefore, we propose a "segment-then-detect" strategy. This approach first segments the region encompassing all teeth and subsequently separates individual teeth from this region. Specifically, inspired by the traditional marker-based watershed transform,[51] known for its effectiveness in segmenting objects with closed contours, we integrate it with deep learning to accurately detect individual teeth in CBCT images. As shown in Figure S3B, the tooth segmentation module includes three nnU-Net[52] sub-networks: the tooth region segmentation sub-network, the tooth boundary and interior segmentation sub-network, and the tooth labeling network. The first two sub-networks generate regions and markers for the marker-based watershed transform, enabling the separation of each tooth. We first process a CBCT image through the tooth region segmentation sub-network to identify the tooth region and obtain accurate contours of all teeth. We then compute the Euclidean distance from the tooth region to create a distance map for watershed processing, segmenting individual teeth. To ensure accurate labels, we use the tooth labeling network followed by a voting strategy to determine the final label.

For maxilla and mandible segmentation, several studies[37,53,54] have proposed the utilization of deep neural networks to analyze alveolar bone structures. However, these studies primarily focus on healthy bone models, resulting in insufficient validation on data pertaining to periodontal disease. In this work, we incorporate a broader dataset that includes cases of periodontal disease. Additionally, we employ nnU-Net as the backbone to generate a 3-channel mask that categorizes regions into background, maxilla, and mandible. During training, both cross-entropy loss and Dice loss are utilized to oversee the segmentation processes of the tooth region, including its boundary and interior, as well as the alveolar bone and tooth labeling processes.

Multimodal data fusion

Our multimodal fusion strategy employs a hierarchical patient-to-tooth cognitive logic, ensuring both global anatomical consistency and local precision required for detailed periodontal assessment, such as accurately calculating gingiva-bone distance at multiple points per tooth. This approach is different from existing methods, such as DDMF,[55] which focuses on point-level registration for general tooth-bone reconstruction but does not achieve the fine-grained accuracy necessary for periodontal metrics. Similarly, TSIM[56] centers on correcting stitching errors in full-arch IOS scans, but its segmentation approach may not be robust enough in scenarios involving common challenges such as metal artifacts, which are prevalent in periodontal cases. By addressing these limitations, our approach leverages both global alignment and local refinement, providing a more precise and reliable solution for periodontal disease diagnosis.

As shown in Figure S3C, we implement a two-stage fusion strategy: 1) reconstruction of the individual tooth and alveolar bone meshes from CBCT image segmentation, and 2) registration of the crown from IOS models to the teeth from CBCT images. Given significant differences in data representation between IOS models and CBCT images, we initially apply the marching cubes algorithm to reconstruct the meshes of teeth and alveolar bone from CBCT images. This algorithm converts the voxel-based segmentation maps into a 3D mesh representation, bridging the data representation gap between two data modalities. For registering the crown to the teeth, we propose a hierarchical registration approach. This process involves coarse-to-fine registration, initially at the patient level and subsequently at the tooth level. Patient-level registration focuses on aligning the global arch position and orientation. We use Principal Component Analysis (PCA) to determine the directions of the three principal directions, which helps calculate rotation and translation parameters. It's important to note that the robustness of PCA in determining principal directions is often limited due to insufficient data on the crown's root. Therefore, we integrate centroid-based information, employing Singular Value Decomposition (SVD) to calculate transformation parameters between the centroids of respective datasets. To further reduce local registration errors, we apply point-to-plane Iterative Closest Point (ICP)[57] both at the patient level and tooth level for more accurate registration.

Digital probing measurement

Figure S3D outlines our approach to obtain gingival and alveolar bone contours for calculating the GBD, and also the process for measuring periodontal staging. After IOS segmentation and multimodal data fusion module, we can calculate digital gingival contours from aligned IOS segmentation results. Initially, we identify vertices along the edge of each aligned crown mesh to define the crown edges. Then, we remove the adjacent points not belonging to the crown for obtaining incomplete gingival edges. In clinics, these missing sections frequently coincide with areas severely affected by periodontal disease, which are challenging to measure clinically due to instrumentation constraints. To overcome this, our method utilizes a spline interpolation approach to arrange points along the gingival edges, completing it into a comprehensive gingival contour consisting of 360 points. Motivated by clinical knowledge, we measure the GBD by aligning the probe with the tooth's growth direction. To ensure accuracy, we first establish the tooth's longitudinal axis using PCA on its vertex data. This axis describes the tooth's shape, guiding our measurements along its length. Given a point on the gingiva contours, we identify its corresponding point on the alveolar bone along this axis, forming bone contours. Post-processing includes removing outliers with over 5mm to enhance measurement reliability. This approach ensures precise and consistent assessment of periodontal disease. Finally, by setting classification thresholds, our system can infer diagnosis of periodontal disease for each tooth based on GBD distances.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses were performed using Python (version 3.9) with matplotlib (v3.5.1), seaborn (v0.12.1), and scipy (v1.13.0).

For the comparison between the IOS segmentation and CBCT image segmentation models (Figures 2C and 2D), we calculated Dice scores as a measure of segmentation accuracy. All reported Dice values represent the mean across samples, with 95% confidence intervals (CI).

For the analysis in the section "digital probing analysis: GBD measurement in clinical space", box plots were generated to visualize (Figures 4A, 4C, and 4F). Box plots show the interquartile range (IQR, 25th–75th percentile), with the median indicated by a line inside the box. Whiskers extend to the most extreme data points within $1.5\times$IQR from Q1 or Q3. Data points beyond this range are plotted as individual outliers.

ADDITIONAL RESOURCES

The study was approved by the Institutional Review Board of the Shanghai Ninth People's Hospital (SH9H-2021-T408-3; trial registration: NCT05513599).

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Supplemental information

PerioAI: A digital system for periodontal disease diagnosis from an intra-oral scan and cone-beam CT image

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Supplemental Information

PerioAI: A digital system for periodontal disease diagnosis from an intra-oral scan and cone-beam CT image

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Supplemental Tables

Table S1. The characteristics of the dataset include modality, pixel spacing, resolution, and imaging protocol. Related to **Figure 2**.

Dataset		Internal set					External set					
		Teeth3DS	CQ-hospital	HZ-hospital	SH-hospital	Clinic#1	Clinic#2	Clinic#3	Clinic#4		SH-hospital	
Modality		IOS	CBCT	CBCT	CBCT	CBCT	IOS	CBCT	IOS	CBCT	IOS	CBCT
Pixel spacing (mm)		-	0.40^3	0.20^3	$(0.25\text{-}0.35)^3$	$(0.20\text{-}0.35)^3$	-	$(0.15\text{-}0.40)^3$	-	$(0.15\text{-}0.40)^3$	-	$(0.15\text{-}0.45)^3$
Resolution		-	(400-640, 400-640, 224-448)	(401, 401, 401)	(428-608, 428-608, 222-440)	(401-672, 401-672, 222-480)	-	(314-900, 314-900, 190-745)	-	(334-1068, 334-1068, 222-868)	-	(334-1068, 334-1068, 223-868)
Image protocals	Manufacturer	-	Sciences Internati-onal	Planmeca	Vatech	Instrume-ntarium Dental	3 Shape	LargeV; Bondent Imaging	3 Shape	Planmeca Trophy; HDXWILL	3 Shape	Planmeca; SOREDEX
	Manufacturer's model	-	17	Promax	XG3D	OP300	TRIOS 3	HighRes3D; Bondream DC3D-1020	TRIOS 3	Viso G5; K9500; Cranex3D	TRIOS 3	Viso G5; SS-X1001DP Plus

Table S2. The dataset's characteristics used in "GBD measurement in clinical space", including patient demographics, disease stage, and other relevant characteristics. Related to **Figure 2** and **Figure 4**.

Dataset		**Total**	**Gingivitis**	**Periodontitis** (n=36)			
				Stage I	Stage II	Stage III	Stage IV
		(n=37)	(n=1)	(n=10)	(n=15)	(n=10)	(n=1)
Demographics							
Gender	Male	18	1	3	7	6	1
	Female	19	0	7	8	4	0
Age (*year*)	Average	39.7	25	37.5	39.3	39.3	54
	18-29	6	1	3	2	0	0
	30-59	29	0	7	12	9	1
	≥60	2	0	0	1	1	0
Smoking status	Non-smokers	29	1	10	11	7	0
	Current	4	0	0	2	1	1
	Former	4	0	0	2	2	0
Clinical parameters							
Number of teeth		26.6	27	26.9	26.7	26.5	23
FMBS		63.5	82.7	60	63.6	63.3	81.2
PPD (*mm*)		3.12	2.77	3.1	3.16	3.13	4.04
CAL (*mm*)		2.06	0.12	1.84	1.99	2.05	5.47

Table S3. Comparison of our method with the other seven methods, i.e., PointNet, PointNet++, Point Cloud Transformer (PCT), TSegNet, SGTNet, DBGANet, and WS-TIS. Related to **Figure 2**.

Model	Internal testing set			External testing set		
	DSC (%)	mIoU (%)	TLA (%)	DSC (%)	mIoU (%)	TLA (%)
PointNet	86.2	79.1	97.2	85.5	77.9	96.4
PointNet++	93.3	89.3	99.1	92.8	88.3	98.6
PCT	95.2	91.8	**99.4**	94.4	90.6	98.8
TSegNet	95.0	91.0	97.8	94.6	90.3	97.7
SGTNet	85.6	76.7	94.9	82.6	73.1	94.9
DBGANet	89.9	85.3	98.9	91.0	86.0	98.2
WS-TIS	93.7	92.6	98.1	94.0	92.2	98.2
Ours	**96.8**	**94.0**	99.3	**96.5**	**93.6**	**98.8**

Table S4. Comparison of our tooth segmentation and alveolar bone segmentation model with the HMG-Net and MWT-Net. Related to **Figure 2**.

Model		Internal testing set		External testing set	
		DSC (%)	ASD (mm)	DSC (%)	ASD (mm)
HMG-Net	Tooth	93.7	0.155	85.4	0.878
	Bone	93.3	0.228	88.8	0.685
MWT-Net	Tooth	75.1	1.734	87.8	0.806
Ours	Tooth	**95.9**	**0.072**	**92.2**	**0.323**
	Bone	**97.2**	**0.072**	**95.9**	**0.238**

Supplemental Figures

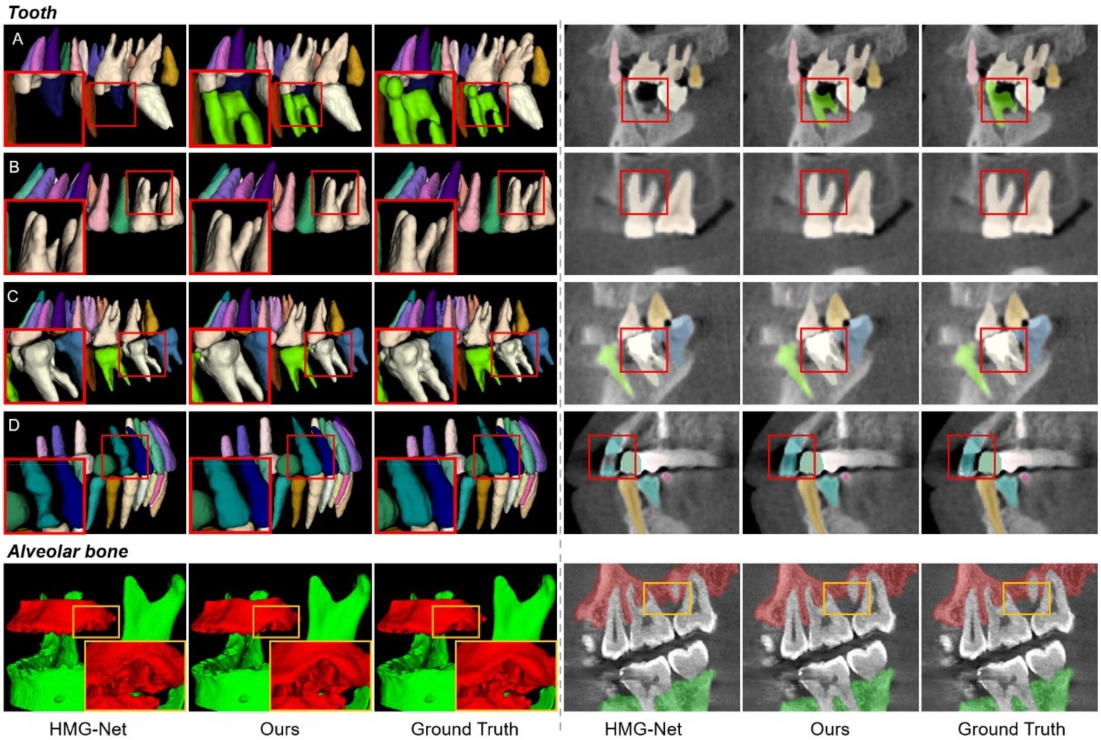

Figure S1. Typical tooth and alveolar bone segmentation results on CBCT images. Related to **Figure 2**.

The missing tooth with caries (A), normal tooth (B), teeth with metal artifacts (C, D), and alveolar bone with bone loss.

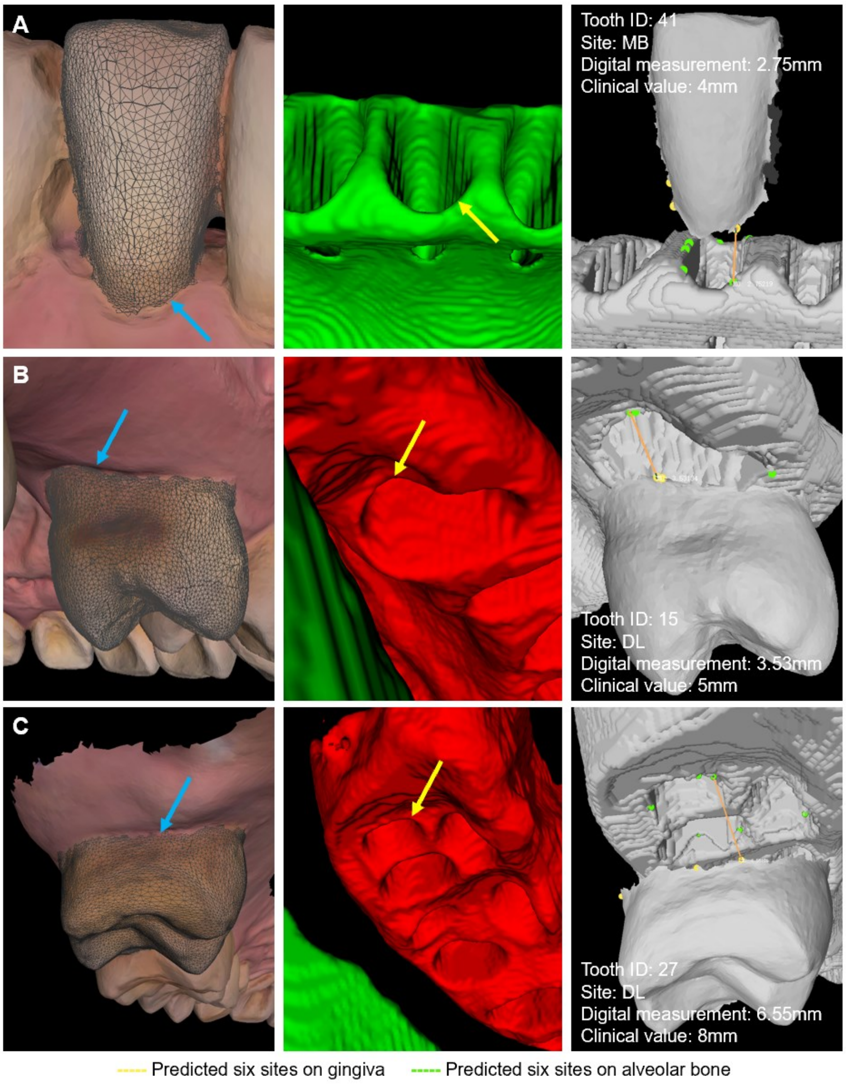

Figure S2. Three typical examples of differences between digital measurements and clinical values for six sites of all teeth. The blue and yellow arrows indicate the corresponding sites in the crown and alveolar bone segmentation results. The yellow and green points mean the predicted six sites on gingiva and bone, respectively. Related to **Figure 4**.

(A) Inconsistent measurement sites may be the reason for measurement values being smaller than clinical values.

(B) Inconsistent methods of digital measurement and clinical measurement may be the reason for measurement differences.

(C) During clinical exploration, probe placement, angle, and subjective judgments relied upon by operators may all be sources of error.

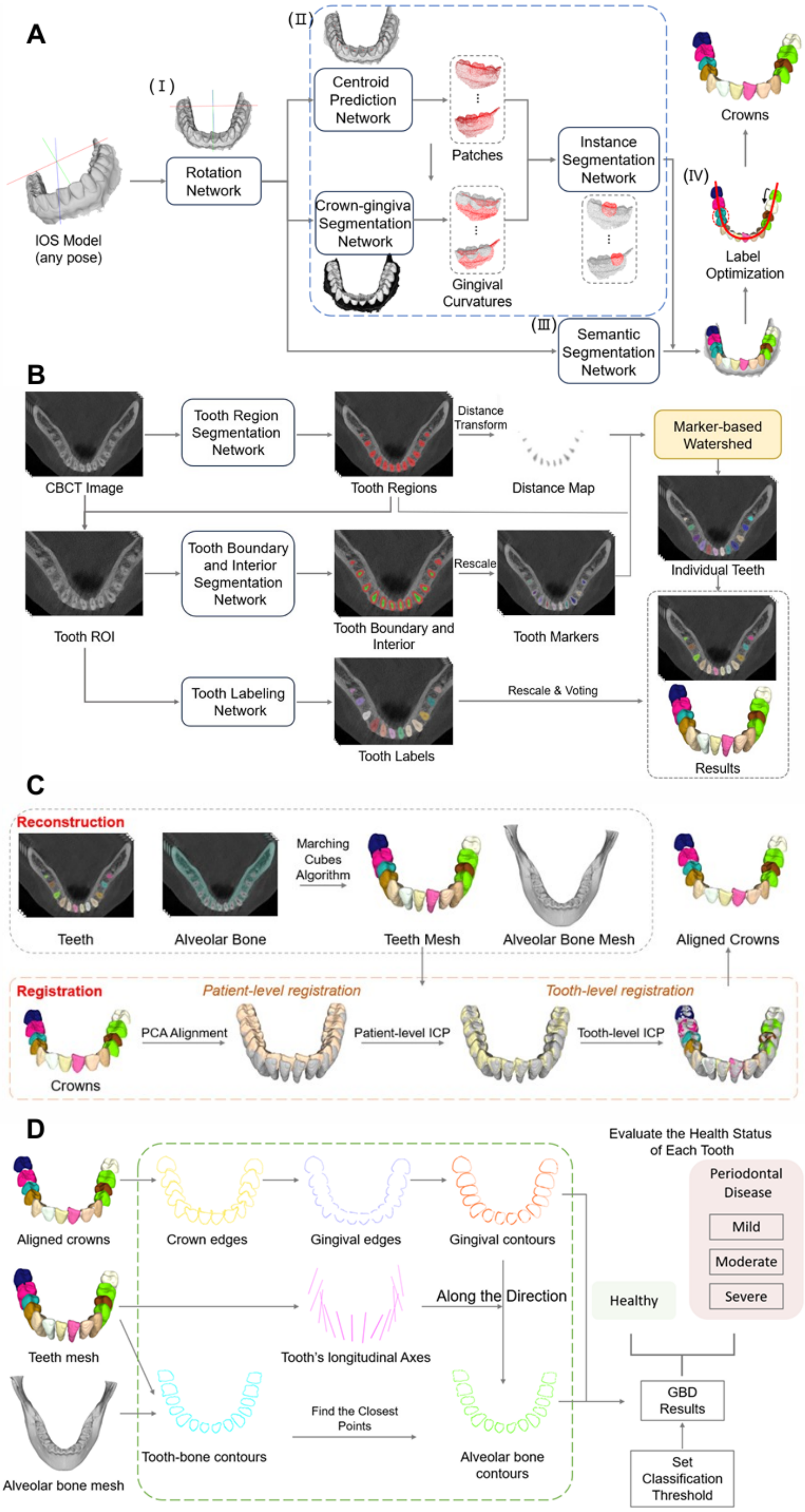

Figure S3. Detail of the digital periodontal diagnosis system. Related to **STAR Methods**.

(A) IOS segmentation can be divided into 1) a rotation network, 2) a tooth instance segmentation module, 3) a tooth semantic segmentation network, and 4) a label optimization module.

(B) Tooth segmentation module on the CBCT image includes three nnU-Net networks and a voting strategy to robustly and accurately segment and classify individual teeth.

(C) The flow chart of the multimodal data fusion module.

(D) Digital probing measurement module.